

GUINAN, China

WMO: 529550

Lat: 35.5867N

Lon: 100.7400E

Elev: 3123

StdP: 69.02

Time Zone: 8.00 (E08)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.7	-18.9	-33.0	0.3	-14.3	-30.9	0.3	-11.8	6.6	-1.2	5.4	-1.7	1.4	160	0.622

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	24.6	12.9	22.7	11.9	21.2	11.0	14.8	21.1	13.9	19.9	13.1	18.3	2.2	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.8	13.6	16.9	12.0	12.9	16.2	11.2	12.2	15.6	53.9	21.2	51.0	20.2	48.2	18.4	18.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.6	4.7	4.0	-24.4	27.8	1.6	1.5	-25.5	28.9	-26.4	29.7	-27.3	30.6	-28.4	31.7		
			-25.1	16.1	1.5	0.9	-26.2	16.8	-27.1	17.4	-28.0	17.9	-29.1	18.6		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-8.8	-4.8	0.1	5.6	8.8	12.1	14.4	14.4	10.3	4.3	-2.6	-7.9		
	DBStd	8.70	3.41	3.91	3.61	3.39	2.92	2.49	2.63	2.79	2.83	3.00	3.18	2.82		
	HDD10.0	2651	584	414	308	138	57	8	1	1	29	176	379	556		
	HDD18.3	5292	842	648	567	383	294	189	124	127	241	433	629	814		
	CDD10.0	409	0	0	1	4	21	69	138	137	38	1	0	0		
	CDD18.3	8	0	0	0	0	0	1	3	4	0	0	0	0		
	CDH23.3	111	0	0	0	0	1	8	44	50	7	0	0	0		
	CDH26.7	8	0	0	0	0	0	0	3	4	1	0	0	0		
(14) Wind	WSAvg	1.8	1.7	1.9	2.1	2.2	2.0	1.7	1.7	1.6	1.5	1.6	1.6	1.6		
(15) Precipitation	PrecAvg	393	2	3	7	20	52	67	88	79	54	17	2	1		
	PrecMax	565	7	9	31	47	95	117	157	150	110	41	11	4		
	PrecMin	243	0	0	0	0	8	33	44	37	17	1	0	0		
	PrecStd	68	2	2	8	14	21	21	25	27	22	10	3	1		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	11.9	17.2	20.7	22.5	24.5	26.7	27.1	23.9	17.9	11.8	6.3		
		MCWB	-1.8	0.2	3.2	5.7	7.8	11.9	14.1	13.8	12.2	6.9	0.9	-2.5		
	2%	DB	4.1	8.7	13.3	18.1	20.0	21.8	24.5	24.7	20.7	15.4	9.4	3.8		
		MCWB	-3.8	-1.3	1.5	4.6	6.9	10.6	13.2	13.2	10.8	5.2	-0.2	-3.9		
	5%	DB	1.7	6.2	10.5	15.9	18.0	20.1	22.6	22.6	18.4	13.4	7.5	1.8		
		MCWB	-5.2	-2.5	0.0	3.8	6.5	9.9	12.5	12.3	9.5	4.1	-1.0	-4.9		
	10%	DB	-0.6	3.7	8.3	13.8	16.1	18.5	20.9	20.7	16.5	11.5	5.4	0.0		
		MCWB	-6.6	-3.7	-1.0	2.9	5.8	9.5	11.9	11.7	8.5	3.2	-2.1	-6.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.8	0.2	3.4	7.2	10.3	15.1	16.6	15.7	13.8	9.5	1.6	-2.3		
		MCDB	7.4	11.0	15.5	15.7	17.2	21.0	23.0	22.9	19.8	15.1	10.4	5.6		
	2%	WB	-3.8	-1.2	1.9	5.6	9.1	13.2	14.9	14.7	12.5	7.1	0.2	-3.8		
		MCDB	3.9	8.2	11.6	14.9	15.7	18.1	21.1	21.2	18.3	11.9	8.4	3.4		
	5%	WB	-5.2	-2.4	0.6	4.6	8.0	11.9	14.0	13.9	11.3	5.5	-0.7	-4.9		
		MCDB	1.4	5.8	9.1	13.6	14.8	16.8	20.1	19.9	16.2	10.8	6.7	1.6		
	10%	WB	-6.5	-3.6	-0.6	3.7	7.0	10.7	13.1	13.1	10.2	4.1	-1.8	-6.0		
		MCDB	-1.0	3.4	7.4	11.9	13.3	15.8	18.3	18.5	14.4	9.5	4.9	-0.3		
(35) Mean Daily Temperature Range	MDBR		16.5	16.0	15.0	13.9	12.1	11.1	11.3	11.9	11.6	13.6	16.0	16.6		
	5% DB	MCDBR	19.3	18.9	18.4	17.4	16.3	15.0	14.6	14.9	15.0	16.0	18.1	18.3		
		MCWBR	13.0	11.4	9.8	7.7	6.6	6.0	5.7	5.5	6.3	7.6	10.1	12.2		
	5% WB	MCDBR	18.7	18.0	16.4	14.4	12.1	11.1	10.8	11.4	11.7	12.8	16.6	17.7		
		MCWBR	12.6	10.9	8.7	6.5	5.4	5.0	4.6	4.5	5.4	6.7	9.4	11.8		
(40) Clear-Sky Solar Irradiance	taub		0.228	0.266	0.320	0.365	0.370	0.361	0.348	0.341	0.316	0.281	0.240	0.217		
	taud		2.499	2.385	2.198	2.093	2.182	2.306	2.406	2.420	2.463	2.491	2.521	2.550		
	Ebn at Noon		1003	985	947	916	913	917	927	930	942	956	978	1000		
	Edh at Noon		79	102	135	159	147	130	117	113	103	90	77	71		
(44) All-Sky Solar Radiation	RadAvg		3.32	4.07	5.20	6.02	6.05	5.85	5.92	5.62	4.64	4.23	3.61	3.15		
	RadStd		0.21	0.20	0.26	0.28	0.32	0.46	0.58	0.43	0.36	0.28	0.16	0.10		

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	+0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page